HSBC BANKING QUERY INTENT ROUTING

A deep learning business application using BANKING77

ISOM5240 - Deep Learning Business Applications with Python

Draft status: Model training and offline test-set experiments are complete. Replace the bracketed items and add deployment screenshots after publishing the app and model.

Project detail	Value
Group	Group 02
Student names and IDs	SHEN, Jia (21158135)
Company and website	HSBC https://www.hsbc.com
Fine-tuned model URL	https://huggingface.co/[owner]/banking77-distilbert
Public Streamlit app URL	https://[app-name].streamlit.app
GitHub repository URL	https://github.com/[owner]/[repository]

Project objective

Classify English online-banking support messages for HSBC into service intents, supporting consistent customer-query routing while evaluating prediction accuracy and inference runtime.

Executive summary

The project proposes two intent-routing approaches for HSBC support on BANKING77: a DistilBERT classifier fine-tuned on the official training data and a pretrained sentence-embedding retrieval baseline. On the complete 3,080-example official test split, the fine-tuned classifier achieved 83.18% accuracy and the retrieval baseline achieved 87.18%. BANKING77 is public, generic banking data rather than HSBC-provided customer messages, so the results do not establish HSBC-specific performance. The Streamlit application supports both pipelines; public deployment remains to be completed.

1. Business Problem and Strategy

Business problem

HSBC online banking support teams may receive short messages about many different services, including card delivery, payments, transfers, cash withdrawals, and identity checks. Manually identifying the service category can slow triage. This application predicts a message's intent to support routing to the relevant service queue; it does not generate banking advice or execute transactions. This is a proposed use case, not an HSBC-approved or HSBC-integrated system.

Strategy

- **Define the task:** map one English customer query to one of the 77 BANKING77 intent labels.

- **Prepare data:** use the dataset's official train/test split and reserve a stratified 10% of training examples for validation.

- **Build two pipelines:** fine-tune one DistilBERT classifier and compare it with a pretrained embedding model plus nearest-neighbor voting.

- **Evaluate consistently:** run both approaches on the same untouched official test examples; measure accuracy and inference time.

- **Apply the result:** present predicted intent(s) in a Streamlit interface and retain the comparison mode for transparent review.

Model workflow

Input query	Tokenize	Pretrained model	Intent decision	Routing category

Pipeline 1 Query - DistilBERT fine-tuned on 77 labels - softmax scores - top intent	**Pipeline 2** Query - pretrained MiniLM embedding - cosine similarity to training examples - top-5 intent vote

Scope: BANKING77 supports single-label intent detection. Sentiment analysis is excluded because this dataset has no verified sentiment labels.

2. Dataset and Pre-processing

BANKING77 is an English, fine-grained banking-domain intent dataset. Each example contains the customer query text and an integer label mapped to a human-readable intent name.

Dataset attribute	Project value
Source	PolyAI/BANKING77 on Hugging Face
Total samples	13,083
Training split	10,003 examples
Official test split	3,080 examples; held out for final evaluation
Number of labels	77 intent classes
Relevant features	text: query string; label: intent class
Language/domain	English; online banking support

Data preparation

- The official 10,003-example training split was divided using a reproducible stratified 90/10 split: 9,002 examples for training and 1,001 for validation.
- The official 3,080-example test split was not used for model training or validation-based checkpoint selection.
- DistilBERT inputs were truncated to 128 tokens and dynamically padded within each batch.
- For retrieval, up to 25 training queries per intent were sampled with seed 42 to create a balanced reference index. Test examples were not included in that index.
- Numeric labels were mapped to the dataset's intent names for readable app predictions and experiment outputs.

Data source and license

Dataset card: huggingface.co/datasets/PolyAI/banking77. The dataset script identifies the dataset license as Creative Commons Attribution 4.0 International; verify attribution and redistribution conditions before submission.

3. Training and Model Design

Fine-tuned classifier

A Hugging Face DistilBERT sequence-classification model was initialized from distilbert/distilbert-base-uncased and fine-tuned to predict the 77 BANKING77 intents. The output configuration stores the label names so predictions are interpretable.

Training setting	Value
Base model	distilbert/distilbert-base-uncased
Epochs	3
Learning rate	2e-5
Batch size	16 train / 32 evaluation
Maximum sequence length	128 tokens
Weight decay	0.01
Random seed	42
Checkpoint selection	Highest validation accuracy

Pretrained retrieval baseline

The second pipeline uses the pretrained sentence-transformers/all-MiniLM-L6-v2 encoder. It mean-pools token embeddings with the attention mask, normalizes the vectors, retrieves the five most similar training examples by cosine similarity, and predicts the most-voted intent. This baseline is not fine-tuned.

Validation results

At the selected best validation checkpoint, the fine-tuned classifier reached **85.21% accuracy** and **0.8302 macro F1** on the 1,001-example validation split. The official test set was reserved for the next experiment stage.

The training and evaluation programs, dependency manifest, Streamlit source, and two Colab-ready notebooks are included in the project folder. The model artifact is saved under `Fine-tuned_Model_files/banking77-distilbert/`.

4. Experiments and Results

Both pipelines were evaluated on the complete 3,080-example official test split. Accuracy and inference runtime are the required comparison measures. Macro F1 is included as a supplementary measure for the 77-class task.

Pipeline	Accuracy	Macro F1	Inference / sample
Fine-tuned DistilBERT	83.18%	0.8224	1.085 ms
Pretrained embedding retrieval	87.18%	0.8714	0.541 ms

Runtime detail	Fine-tuned classifier	Embedding retrieval
Test-set inference time	3.343 s	1.666 s
Model/index setup time	Not recorded in this run	2.065 s (includes reference encoding)
Evaluation examples	3,080	3,080

Interpretation

- On this test set, retrieval exceeded the fine-tuned classifier by 3.99 percentage points in accuracy and 0.0491 in macro F1.
- Retrieval also had lower per-query inference time, excluding its one-time model and reference-index setup cost.
- The reported timings were measured locally on Apple MPS; they should not be treated as Streamlit Cloud latency estimates.

Model-selection note: The current measurements favor retrieval. The final deployment choice should be confirmed against the course requirement to keep the app model aligned with the fine-tuned artifact. The app source includes both pipelines and a side-by-side mode.

The generated Excel workbook contains the comparison table and all test predictions: `results/banking77_comparison.xlsx`.

5. Business Application and Deployment

Application behavior

The Streamlit interface accepts an English banking-support message and offers the fine-tuned classifier, pretrained retrieval, or side-by-side comparison. It displays the predicted intent and alternative intents. Classifier scores are model probabilities; retrieval scores are mean cosine similarities and are not calibrated probabilities.

Cloud deployment status

The app code, requirements, Community Cloud configuration, and deployment instructions are prepared. The app has **not yet been published**, and no public app, model, or GitHub URL is claimed in this draft. Before submission, publish the fine-tuned model on Hugging Face, deploy the app from GitHub, set `BANKING77_MODEL_ID` in Streamlit Community Cloud secrets, and test the deployed app.

Required evidence	Insert before final submission
App URL	[Public Streamlit Cloud URL]
Model URL	[Public Hugging Face fine-tuned model URL]
GitHub URL	[Repository linked to the deployed Streamlit app]

Deployment screenshots

[Insert screenshot 1: Streamlit app input and prediction result. Include a short caption and confirm the deployed public URL.]
[Insert screenshot 2: comparison mode showing the fine-tuned classifier and retrieval predictions for the same query.]

Application accuracy currently refers to offline evaluation of the app's fine-tuned classifier path on the held-out test data. Accuracy and operational behavior on the public Streamlit deployment remain to be verified.

6. Conclusion, Limitations, and Sources

Conclusion

The project demonstrates a practical banking-query intent-routing application using Hugging Face models. The fine-tuned DistilBERT classifier achieved 83.18% test accuracy, while the pretrained retrieval baseline achieved 87.18%. These results support the feasibility of automated intent triage on BANKING77, but they do not establish a reduction in real-world handling time or customer effort.

Limitations and next steps

- BANKING77 is English-only and represents a fixed set of 77 banking intents; it is not HSBC-provided data, so performance on HSBC customer messages, other languages, and newly emerging intents is unknown.

- Test results are offline benchmark results. Public Streamlit availability, cloud latency, and real agent-routing outcomes have not yet been measured.

- Some intents are semantically similar. Review the exported per-example predictions and confusion patterns before operational use.

- Complete the bracketed team/company/URL fields, upload the exact trained model, deploy the app, capture screenshots, and rerun the final test from the deployed application path before treating this as the final report.

Sources

1. PolyAI, BANKING77 dataset card: https://huggingface.co/datasets/PolyAI/banking77.

2. Casanueva et al., *Efficient Intent Detection with Dual Sentence Encoders*, 2020: https://arxiv.org/abs/2003.04807.

3. Hugging Face base checkpoint: distilbert/distilbert-base-uncased.

4. Hugging Face embedding checkpoint: sentence-transformers/all-MiniLM-L6-v2.

5. Streamlit Community Cloud deployment and secrets guides: deployment and secrets management.

6. HSBC corporate website: https://www.hsbc.com.